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2) octane C_8H_{18} $\phi = 0.5$ $T_{reat} = 298K$

$P = 1$ atm

$$C_8H_{18} + a(O_2 + 3.76N_2) \rightarrow xCO_2 + \frac{y}{2}H_2O + 3.76 \cdot a \cdot N_2$$

when $x = 8$, $y = 18$

$a\ O_2 \rightarrow 16\ O + 9\ O$

$2a = 25$

$a = 12.5$

$\phi = \frac{(f/a)_{actual}}{(f/a)_{stoich}} \Rightarrow 0.5 = \frac{(f/a)_{actual}}{1/12.5}$

$\boxed{(f/a)_{actual} = \frac{1}{25}}$

fuel lean = extra oxidizer

$$\boxed{C_8H_{18} + 25(O_2 + 3.76N_2) \rightarrow 8CO_2 + 9H_2O + 12.5\,O_2 + 94N_2}$$

$50 \cdot O \rightarrow 2x + \frac{y}{2} + 12.5\,O_2$ ✓

~~chemical~~ H balance

$$\sum N(H(T_f) - H_{298})_p = \sum N_R \Delta h^\circ_{298_R} - \sum N_P \Delta h^\circ_{298_P}$$

RHS:

N_2 O_2 talk Ax

$1(-250)_{C_8H_{18}} + 0 + 0 - 8(-393) - 9(-242)$

$= 5072$ kJ/mol

guess $T_f = 1500$ K

$\sum N(H(T_f) - H_{298})_p = 8(61.7) + 9(48) + 12.5(40.6) + 38.4(94)$

$= 5044.56$ kJ/mol

$5044 \approx 5072$ thus $\boxed{1500K < T_f < 1600K}$

looking at the fig on p.25, $\boxed{H/C = 18/8 = 2.25}$, $E = \frac{+250}{(12.8 + 1.18)} = +2.2$ kJ/g $= \boxed{+0.51 \text{ kcal/g}}$

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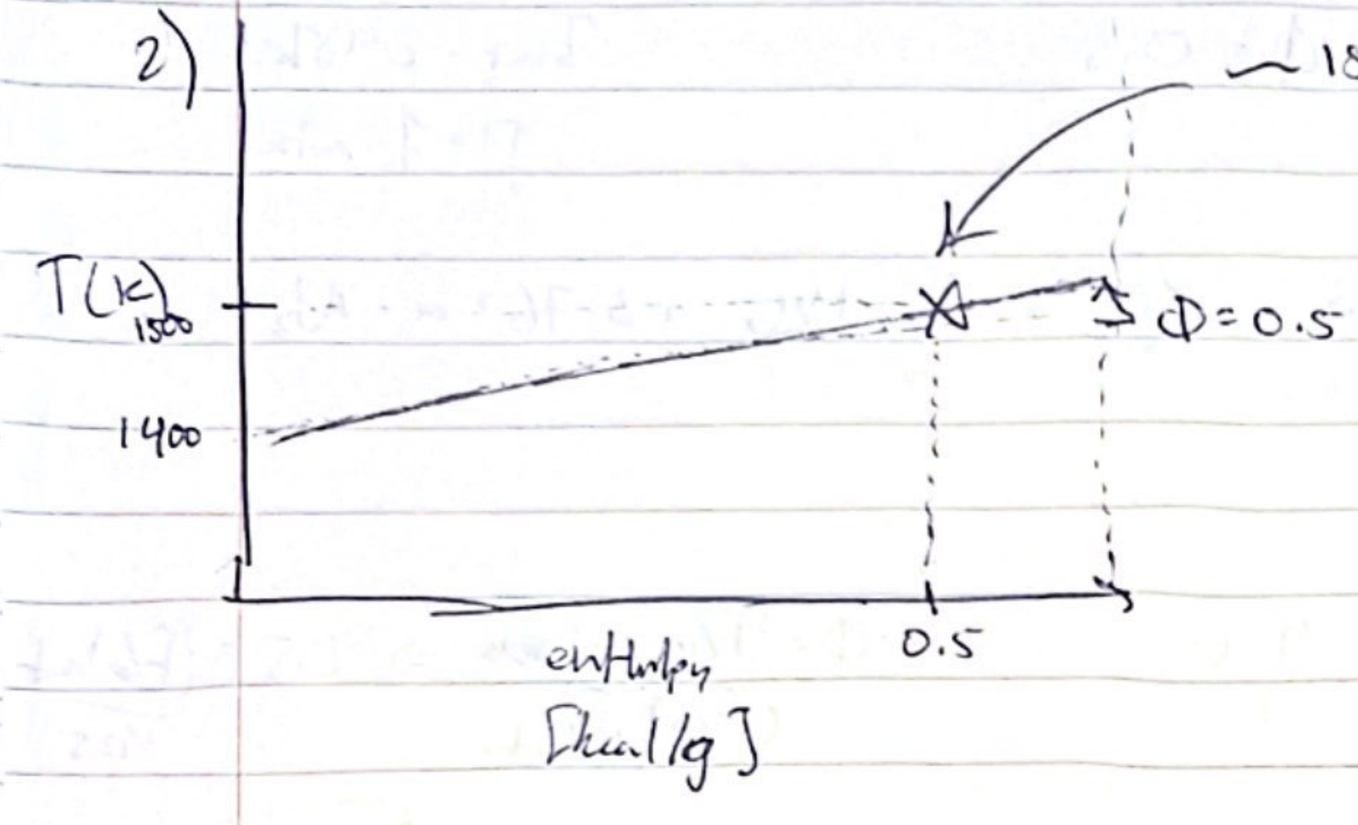
2)



~1500 K, agrees with calculation ✓

H/C = 2.25

also, LEA gives 1504 K ✓

5) $p = 20$ atm $\quad T_f = ?$

Compound	Mole fraction
H_2O	0.493
H_2	0.498
O_2	0
O	0
H	0.02
OH	0.005

$K_{p,f} \Rightarrow \quad \frac{1}{2} H_2 \rightleftharpoons H$

$$K_{p,f} = \frac{0.02}{0.498^{1/2}} \cdot \left(\frac{20}{.493 + .498 + .02 + .005} \right)^{0.5}$$

$$= 0.1257$$

$$\log_{10}(0.1257) = -0.901$$

from table A2, p 599

$\log_{10} k = -0.901$ yields: $\boxed{2900\text{ K} < T_f < 3000\text{ K}}$

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11) $p = 20$ atm

$$C_2H_4O(l) \rightarrow aCO + bCH_4 + cH_2 + dC_2H_4$$

C: $2 = a + b + 2d$
H: $4 = 4b + 2c + 4d$
O: $\boxed{1 = a}$

$\downarrow$

$1 = b + 2d$
$2 = 2b + c + 2d$

equilibrium: $2CH_4 \rightleftharpoons 2H_2 + C_2H_4$

$K_p = ?$

$$K_p = \frac{n_{H_2}^2 \cdot n_{C_2H_4}}{n_{CH_4}^2} \cdot \left(\frac{20}{\Sigma n}\right)^{2+1-2=1}$$

$$K_p = \frac{c^2 \cdot d}{b^2} \cdot \left(\frac{20}{\text{carbon}}\right)^1$$

$$\Sigma N_p (H(T_f) - H_{298})_p = \Sigma N_R \Delta h^\circ_{298\,R} - \Sigma N_p \Delta h^\circ_{298\,p}$$

from tables

RHS: $1(-78) - a(-110) - b(-74.8) - c(0) - d(52)$
: $-78 + 110a + 75b - 52d$

using CEA:

$\boxed{T_f = 1273\,K}$

compound	mole fraction:
~~etanol~~ CH_4	0.416
CO	0.483
H_2	0.06692
C_2H_4	0.03346

check T_f using mole fraction $K_p \Rightarrow \log_{10}\left[\left(\frac{0.664^2 \cdot 0.0335}{0.416}\right)\left(\frac{20}{2}\right)^1\right]$

$\log_{10}(K_p) = -1.76$
using table A_2, p590;

T, total mol $\Rightarrow 1 = 0.416T + 0.035T(2)$
$T = 2.071$ mol, thus:

1100 K < T_f < 1200 K ✓

$$\boxed{C_2H_4O(l) \rightarrow CO + 0.86\,CH_4 + 0.139\,H_2 + 0.07\,C_2H_4}$$

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12) $N_2H_4(l) \rightarrow aN_2 + bH_2 + cNH_3$ $\qquad P = 100$ atm

N: $2 = 2a + c$

H: $4 = 2b + 3c$

Kp: $\frac{1}{2}N_2 + \frac{3}{2}H_2 \rightleftharpoons NH_3$

$$K_p = \frac{n_{NH_3}}{(n_{N_2})^{1/2}(n_{H_2})^{3/2}}\left(\frac{100}{\Sigma n}\right)^{1-\frac{1}{2}-\frac{3}{2}=-1}$$

$$K_p = \frac{c}{a^{1/2}b^{3/2}}\left(\frac{100}{a+b+c}\right)^{-1/2}$$

$$\Sigma N_P(H(T_f) - H_{298})_P = \Sigma N_R h^\circ_{298\,R} - \Sigma N_P h^\circ_{298\,P}$$

RHS: $2(50.67) - a(0) - b(0) - c(-45.9) = a(H(T_f) - H_{298})_{N_2} + b(H(T_f) - H_{298})_{H_2} + c(H(T_f) - H_{298})_{NH_3}$

From CEA:

$T_f = 918$ K

compound	mole fraction
NH_3	0.03114
H_2	0.6407
N_2	0.328

check using k_p, mole fractions $\Rightarrow$ $\log_{10}(k_p) = \log_{10}\left(\frac{0.03}{(0.328)^{1/2}(0.641)^{3/2}}\left(\frac{100}{1}\right)^{-1/2}\right)$

$= -2.99$

from table A2, p607

$\log_{10} k_p = -2.99$ indicates $\boxed{900\,K < T_f < 1000\,K}$ ✓

~~[illegible]~~! T_f, total mol $\Rightarrow$ $2 = 2(0.328)T_f + 0.03114$

$T = 2.91$

thus:

$$\boxed{N_2H_4(l) \rightarrow 0.95\,N_2 + 1.863\,H_2 + 0.091\,NH_3}$$

Stoich: $1 O_2 + 2 H_2 \Rightarrow 2 H_2O$

F/A = 2

thus $\boxed{\phi = 5/2 = 2.5}$

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13) $O_2 + 5H_2 \rightarrow aH_2O + bH_2 + cH$

p = 1 atm

$T_i = 298K$

O: $2 = a$

H: $10 = 2a + 2b + c$

$6 = 2b + c$

$a, b, c, T_f = ?$

K_p: $\frac{1}{2}H_2 \rightleftharpoons H$

$$K_p = \frac{n_H}{(n_{H_2})^{1/2}}\left(\frac{1}{\Sigma n}\right)^{1-\frac{1}{2}=\frac{1}{2}}$$

$$K_p = \frac{c}{b^{1/2}} \cdot \left(\frac{1}{a+b+c}\right)^{1/2}$$

$$\Sigma N_p(H(T_f) - H_{298})_p = \Sigma N_p \Delta h^\circ_{298,p} - \Sigma N_p \Delta h^\circ_{298,p}$$

RHS: $0 + 0 - a(-241.8) + b(0) - c(218)$

$= 241.8a - 218c$

from CEA:

$\boxed{T_f = 2647\ K}$ ~~mole fraction~~ sorry!!

~~H 0.00125~~

~~H_2O 0.8263~~

→ plug into K_p ←

comp	mole fraction
H	0.03486
H_2	0.57211
H_2O	0.39303

$\log_{10}\left(\frac{0.03486}{.5721^{1/2}}\right) = -1.34$, using table A2, p. 599

$2600K < T_f < 2700K$ ✓

$6 = 2(0.57211)T + 0.03486(T) \rightarrow T_{moles} = 5.088$ mol

$$\boxed{O_2 + 5H_2 \rightarrow 1.99H_2O + 2.91H_2 + 0.177H}$$

a b c

plug into EHS, 449 kJ ≈ 442 ✓

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14) N_2O_4 + UDMH ($C_2H_8N_2$) o/f = 2

$\boxed{\text{see next page for all equations + } T_f}$

p = 67 atm

products: N_2, CO_2, H_2O, CO, H_2, O, H, OH, NO

ΔH_{f298} N_2O_4 = −21 kJ/kmol

$\Delta H_{f,298}$ UDMH = 53.2 kJ/kmol

9 unknowns

$$x N_2O_4 + y C_2H_8N_2 \rightarrow a N_2 + b CO_2 + c H_2O + d CO + e H_2 + f O + g H + h OH + i NO$$

① N: $2x + 2y = 2a + i$

② O: $4x = 2b + c + f + h + i$

③ C: $2y = b + d$

④ H: $8y = 2c + 2e + g + h$

Kp: $\frac{1}{2}H_2 \Leftrightarrow H$

⑥ $K_{p_1} = \frac{g}{e^{1/2}} \cdot \left(\frac{67}{\Sigma n}\right)^{1/2}$

⑤ $\Sigma N_r (H(T_f) - H_{298}) = \Sigma N_R \Delta^o h_{298\,R} - \Sigma N_P \Delta^o h_{298\,P}$

RHS: $x(-21) + y(53.2) - a(0) - b(-393) - c(-242) - d(-110) - e(0) - f(249)$

$- g(218) - h(39) - i(90)$

using CEA: $\boxed{T_f = 3293\ K}$

need 3 more Kp eqns:

$H + \frac{1}{2}OH \Leftrightarrow 2H_2O$

⑦ $K_p = \frac{i^2}{f^2 g} \left(\frac{67}{\Sigma n}\right)^{2-2-1=-1}$

$CO + O \rightleftharpoons CO_2$

⑧ $K_p = \frac{b}{f\,d} \left(\frac{67}{\Sigma n}\right)^{1-1-1=-1}$

$\boxed{\text{See next page for list of all eqns neatly}}$

⑨ $H_2 + OH \rightleftharpoons 2H_2O \Rightarrow K_p = \frac{c^2}{e\,h^2} \left(\frac{67}{\Sigma n}\right)^{2-2-1=-1}$

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14) 9 equations to solve

(1) N: $2x + 2y = 2a + i$

(2) O: $4x = 2b + c + f + h + i$

(3) C: $2y = b + d$

(4) H: $8y = 2c + 2e + g + h$

where x, y found using O/F ratio, MW ⇒ $MW_{mixture} = 92\ g/mol$
$MW_{N2O4} = 60\ g/mol$ ☆

(5) $k_{p_1} = \frac{g}{e^{1/2}} \left(\frac{67}{\Sigma n}\right)^{1/2}$

(6) $k_{p_2} = \frac{i^2}{f^2 a} \left(\frac{67}{\Sigma n}\right)^{-1}$

(7) $k_{p_3} = \frac{b}{fd} \left(\frac{67}{\Sigma n}\right)^{-1}$

(8) $k_{p_4} = \frac{c^2}{eh^2} \left(\frac{67}{\Sigma n}\right)^{-1}$

(9) $\Sigma N_r (H(T_f) - H_{298K}) =$

$x(-21)\ y(53.2) - \cancel{0}\ a - b(-393) - c(-242) - d(-110) - \cancel{0}\ e - f(249) - g(218)$
$- h(39) - i(90)$

$\boxed{\text{using CEA, } T_f = 3293\ K}$

☆

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15) using CEA to run the four cases:

pure O_2, $0.21\ O_2 + 0.79\ N_2$, $0.21\ O_2 + 0.79\ Ar$, $0.21\ O_2 + 0.79\ CO_2$

ran $p = 1$ atm, C_8H_{18} (l), $T_i = 298$ K, $\phi = 1$

$T_{f,O_2} = 3097$ K

$T_{f,air} = 2261$ K

$T_{f,O_2+Ar} = 2553$ K

$T_{f,O_2+CO_2} = 1812$ K

highest to lowest:

- pure O_2 (0.21 mol)
- $0.21\ O_2 + 0.79\ Ar$
- Air
- $0.21\ O_2 + 0.79\ CO_2$

Some of this can be attributed to the weight of each oxidizer and the heat capacity of every $+0.79 \ldots$ component

- CO_2 will have a higher Δh° than N_2 for example:

at $T = 3000$ K:

$\Delta h^\circ_{N_2} = 93$ kJ/mol

$\Delta h^\circ_{CO_2} = 153$ kJ/mol